

Friday, July 17th

Focus:

Jamshed Karimnazarov

This week's plan

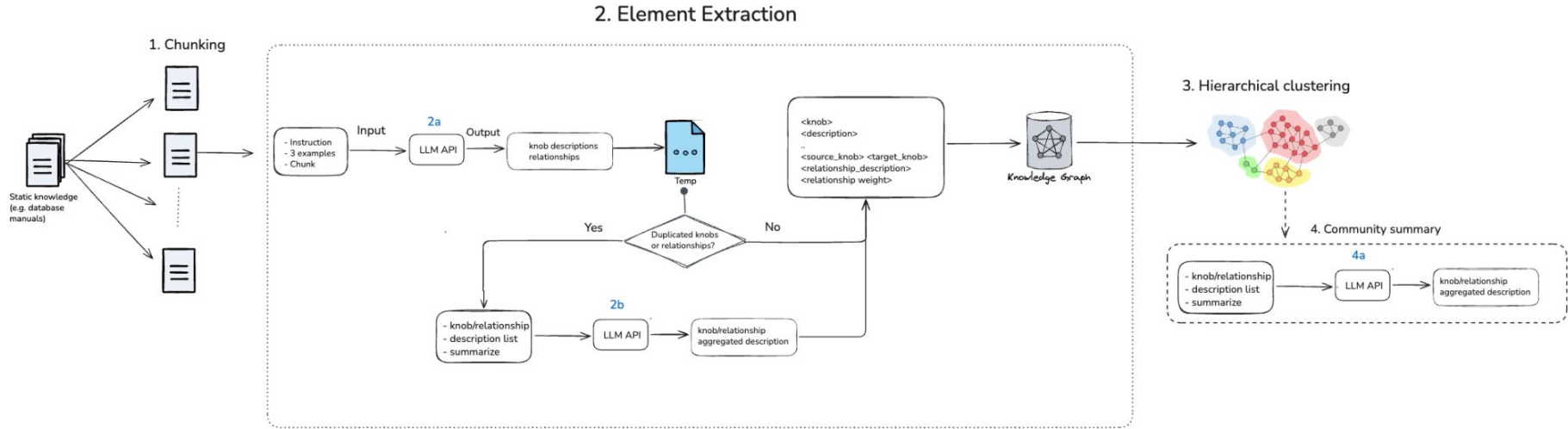
- 01 Count the number of relationships in the input file
- 02 Calculate Recall and F1 score for my solution
- 02 Come up with a solution to filter the input data

Table of contents

1. Component 1: Knowledge Base construction— Slide 4
2. Input data preparation— Slides 5
3. Issue 1: Hallucination in LLM- 2a and Solution— Slide 6-9
4. Issue 2: Filtering DBMS manual and solution— Slide 10-11
6. Next steps— Slide 11

Total: 11 Slides

Component 1: Knowledge Base construction

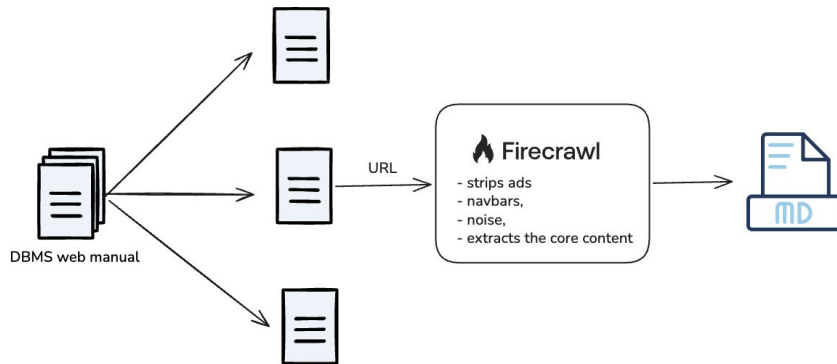


Input data preparation

Select 3 pages:

- [InnoDB parameters](#)
- [Internal Temporary Table](#)
- [What the MySQL Upgrade Process Upgrades](#)

Total: 184 knobs



Issue 1: RABBIT hallucinations in LLM 2a component

Goal: Extract from chunks

- All knobs + descriptions
- All knob relationships + description

Hallucination type: Entity and relationship error

	Knob extraction	Relationship extraction
Precision	0.17	0
Recall	0.07	0
F1 Score	0.10	0

Solution for issue 1

Focus on LLM-2a component where most of the hallucination happens

Cause of hallucination:

- Default GraphRAG prompt
- Default entity types

Prompt Template

```
-Goal-
Given a text document that is potentially relevant to this activity, identify the knobs and the relationships among the
identified knobs.

Database: {dbms_name} {dbms_version}

Knob definition: {knob_definition}

DBMS-specific terminology:
In {dbms_name}, knobs may be referred to as: {knob_terms}.

-Steps-
1. Identify all the knobs. For each knob, extract the following information:
- knob_name: Name of the knob, capitalized
- knob_description: Description of the knob in 1-3 sentences. Include any default values or ranges if mentioned in the text.
If description is not included, keep it empty.
Format each knob as ("knob"{tuple_delimiter}<knob_name>{tuple_delimiter}<knob_description>)

2. From the knobs identified in step 1, identify all pairs of (source_knob, target_knob) that are *clearly related* to each
other.
For each pair of related knobs, extract the following information:
- source_knob: name of the source knob, as identified in step 1
- target_knob: name of the target knob, as identified in step 1
- relationship_description: explanation as to why you think the source knob and the target knob are related to each other
Format each relationship as ("relationship"{tuple_delimiter}<source_knob>{tuple_delimiter}<target_knob>{tuple_delimiter}
<relationship_description>)

3. Return output in English as a single list of all the knobs and relationships identified in steps 1 and 2. Use **
{record_delimiter}** as the list delimiter.

4. When finished, output {completion_delimiter}

#####
-Examples-
#####
{examples}

#####
-Real Data-
#####
Text: {input_text}
#####
Output:
```

My solution: Element extraction - LLM 2a

Knobs

- Precision: **0.74** (0.17)
- Recall: **0.88** (0.07)
- F1-score: **0.80** (0.10)

Relationships

- Precision: **0.9** (0.75)
- Recall: **0.75** (0.00)
- F1-score: **0.77** (0.00)

Issue 2: Filtering the DBMS manual and chunking

Key Challenges

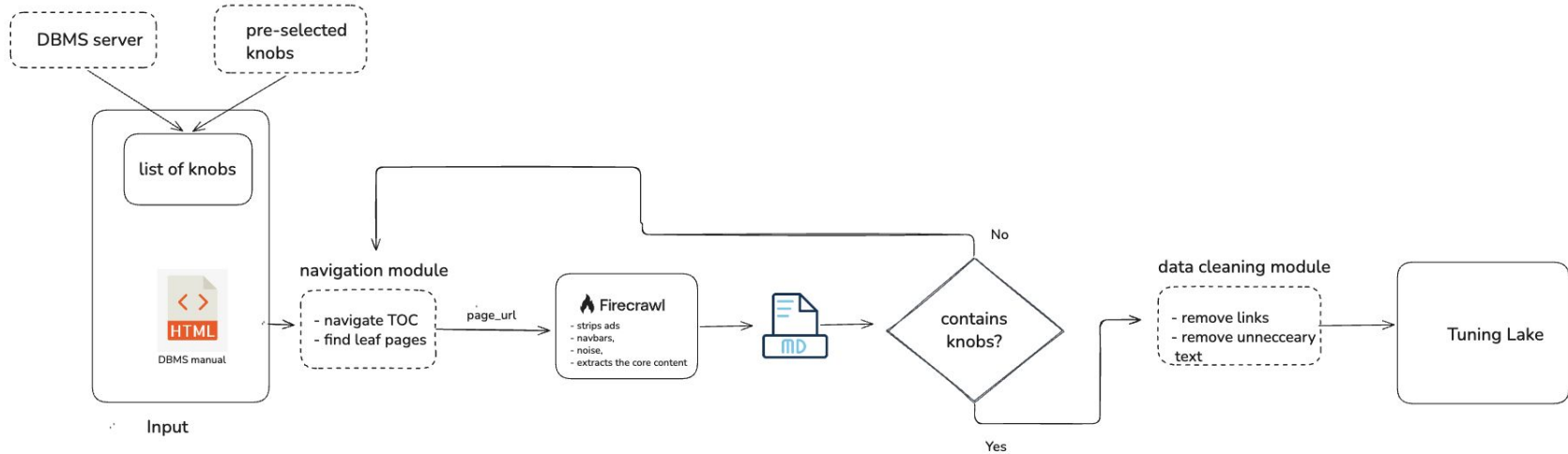
- RABBIT uses crafted input data in their experiments
- Increased hallucination risk with irrelevant pages

Token Usage Concerns

- MySQL manual spans 6,000 pages'
- Just 3 selected pages can require 200K tokens

Goal: Filter the DBMS manual to exclude irrelevant sections

Solution for issue 2: DBMS manual pruning module



Next steps

01 Chunking the documents

02 LLM-2b (summarization component)

03 Constrained decoding guardrails